METACHRONAL CASCADE FLAPPER

Wave Function and Maneuver Control Guide

Eight-Wing Staggered Cascade Ornithopter — Drone Prototype

How the metachronal wave works and how it is modified to achieve every flight mode from stable hover to aerobatic maneuvers

Document scope	Complete explanation of metachronal wave operation and all maneuver control strategies
Vehicle	Eight-wing staggered cascade drone, 1 Hz flap, 3 DoF per wing root
Maneuvers covered	Hover, forward, reverse, lateral, climb, descent, turns, rolls, loops, split-S, Immelmann, knife-edge, inverted, snap roll, spin, hammerhead, and emergency recovery
Control axes	Three independent mechanisms per axis — wave phase, stroke amplitude, angle of attack differential, and sweep angle differential
Key principle	Every maneuver is achieved by modifying wave parameters in software. No mechanical reconfiguration is required between flight modes.

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CHAPTER 1 — HOW THE METACHRONAL WAVE WORKS

1.1 The Fundamental Concept

The metachronal wave is a traveling phase offset applied sequentially to a row of flapping wings. Rather than all wings beating in unison, each wing begins its power stroke a fixed time interval after the wing immediately in front of it. The result is a smooth wave of motion that travels from the frontmost wing to the rearmost wing on each side of the fuselage — identical in principle to the wave that travels along a centipede's legs or down the fin of a cuttlefish.

At the default setting of 90-degree phase spacing between adjacent wings, with a 1 Hz flap frequency, the timing across the four-wing stack on each side is:

Wing	Phase	Delay	Description
Wing 1 — front	Phase 0 deg	0 ms	Power stroke begins at cycle start
Wing 2	Phase 90 deg	250 ms	Power stroke begins 250 ms later
Wing 3	Phase 180 deg	500 ms	Power stroke begins 500 ms later
Wing 4 — rear	Phase 270 deg	750 ms	Power stroke begins 750 ms later

At any given instant during the cycle, the four wings are evenly distributed around the full 360-degree cycle. One wing is always near the start of its power stroke, one is at mid power stroke, one is at stroke reversal, and one is in mid recovery. Lift generation is therefore nearly continuous — there is no moment when all wings are simultaneously in recovery and generating no lift.

1.2 The Cycle in Detail

Each wing executes three simultaneous motions within every 1,000 ms cycle:

- STROKE (phi axis): The wing sweeps through a 120-degree arc — 60 degrees either side of the neutral position. This is a sinusoidal motion. The power stroke is the downward sweep; the recovery stroke is the upward return.

• PITCH (psi axis): The angle of attack follows a smoothed square wave, not a sinusoid. During the power stroke the wing holds at 35–42 degrees AoA for maximum lift. At stroke reversal the pitch servo executes a rapid flip to near-zero degrees — feathering the wing so the recovery stroke generates minimal drag.

• SWEEP (chi axis): The wing sweeps progressively forward during the power stroke (maximum span, maximum lift) and progressively aft during recovery (reduced area, minimum drag). Full aft position equals the stowed position.

The combination of these three motions produces a wingtip path that traces a crescent or comma shape in three-dimensional space — exactly the path observed in the wingtips of all slow-flying vertebrate birds.

1.3 Why the Wave Produces Stable Lift

The metachronal wave solves the fundamental problem of all flapping wing systems: the gap between power strokes. In a conventional two-wing design, both wings are simultaneously in recovery for half of every cycle. During that half-cycle the vehicle is falling. The metachronal wave eliminates this gap entirely. With 90-degree spacing across four wings per side, at least one wing per side is always within the productive first quarter of its power stroke at every moment. The fuselage experiences near-constant lift with only small ripple oscillations — similar to the smoothing effect of multiple cylinders in an engine versus a single-cylinder design.

KEY PRINCIPLE: The metachronal wave is not just a timing strategy — it is the primary reason this architecture can hover. Without it, a 1 Hz flapping system would fall between strokes. With it, lift is effectively continuous.

1.4 The Wave as a Control Surface

Every flight maneuver is achieved by modifying one or more parameters of the metachronal wave. The wave has six independently adjustable parameters, each of which produces a distinct aerodynamic effect:

Parameter	Definition	Aerodynamic Effect
Phase spacing	Degrees between adjacent wings	Changes wave travel speed and interference pattern
Wave direction	Front-to-back or back-to-front	Reverses thrust direction — forward or reverse flight

Parameter	Definition	Aerodynamic Effect
Phase asymmetry	Different spacing left vs right	Generates yaw torque — turns the aircraft
Amplitude	Stroke arc in degrees	Controls total lift and thrust magnitude
Amplitude asymmetry	Different amplitude left vs right	Generates roll moment
AoA profile	Power and recovery AoA values	Controls lift-to-drag ratio and both-stroke lift capability
Sweep schedule	Forward/aft sweep timing within cycle	Controls effective span, thrust vector, and stow depth
Fore/aft AoA bias	Different AoA front wings vs rear wings	Generates pitch moment — nose up or nose down

CHAPTER 2 — THE THREE CONTROL MECHANISMS

Every control axis — roll, pitch, yaw, and altitude — has three independent mechanisms that can act simultaneously or independently. This redundancy means that if one mechanism saturates or fails, the other two continue providing control authority. It also means that maneuvers can be blended with great precision by mixing contributions from all three mechanisms.

ROLL CONTROL	
Mechanism 1	Differential stroke amplitude — left wings stroke further than right (or vice versa), producing more lift on one side
Mechanism 2	Differential AoA — left wings hold higher angle of attack during power stroke, generating more lift coefficient on that side
Mechanism 3	Differential sweep angle — left wings sweep further forward during power stroke, increasing effective span and CL on that side

PITCH CONTROL	
Mechanism 1	Fore/aft AoA bias — front wings (W1, W2) hold higher AoA than rear wings (W3, W4) to pitch nose down; reverse for nose up
Mechanism 2	Stroke plane tilt — all wings' stroke planes tilted forward or backward, shifting the thrust vector fore or aft
Mechanism 3	Fore/aft amplitude bias — front wings stroke further than rear wings, shifting the centre of lift forward

YAW CONTROL	
Mechanism 1	Differential wave phase — left side wave phase shifted forward, right side shifted back, creating asymmetric thrust timing that generates yaw torque

Mechanism 2	Wave direction asymmetry — left side wave travels front-to-back while right side travels back-to-front, generating opposing thrust vectors
Mechanism 3	Differential sweep direction — one side sweeps forward (more drag), other side sweeps aft (less drag), creating yaw torque from drag asymmetry

ALTITUDE / THROTTLE	
Mechanism 1	Phase compression — phase spacing reduced from 90 degrees toward 0 degrees (more in-phase), increasing total instantaneous lift
Mechanism 2	Amplitude increase — all wings stroke through wider arc, generating more lift per cycle at same frequency
Mechanism 3	Both-stroke AoA — recovery AoA raised from near-zero to 20+ degrees, making the recovery stroke generate positive lift rather than just recovering

CHAPTER 3 — FUNDAMENTAL FLIGHT MODES

STATIONARY HOVER	
Mechanisms used	Counter-phase wave (180 degrees between front pair and rear pair). Symmetric amplitude left and right. All wing AoA at hover setting (35 degrees power, 5 degrees recovery). Sweep schedule: moderate forward on downstroke, moderate aft on recovery.
Wave parameters	Phase: 0/180/0/180 degrees (W1/W2/W3/W4). Amplitude: 60 degrees all wings. Left-right: fully symmetric. AoA: 35/5 degrees all wings.
Expected result	Near-zero body oscillation. Minimum power consumption for sustained hover. Equivalent to counter-rotating helicopter rotor pairs cancelling torque.

FORWARD FLIGHT (CRUISE)	
Mechanisms used	Metachronal wave traveling front-to-back (0/90/180/270 degrees). Stroke plane tilted forward 15–25 degrees by biasing all wing stroke centres forward. Forward sweep increased during power stroke to maximise span.
Wave parameters	Phase: 0/90/180/270 degrees. Stroke plane: 15–25 degree forward tilt. Sweep: 30 degrees forward at mid-stroke. AoA: 28–32 degrees (reduced from hover).
Expected result	Net thrust vector has forward component. Body pitches slightly nose-down. Estimated cruise: 18–28 km/h. Wings cycle efficiently at reduced AoA.

REVERSE FLIGHT	
Mechanisms used	Wave direction reversed: back-to-front (270/180/90/0 degrees). Rear wings now lead the wave. Stroke plane tilted aft. No mechanical reversal required — pure software parameter change. Sweep schedule also reversed.

Wave parameters	Phase: 270/180/90/0 degrees (W1/W2/W3/W4 — W4 now leads). Stroke plane: 15 degree aft tilt. Sweep: forward sweep now beneficial on what was the recovery stroke.
Expected result	Rearward thrust vector. Estimated reverse speed: 0–8 km/h. Less efficient than forward flight — cascade interaction is not optimised for this direction but is fully functional.

VERTICAL CLIMB	
Mechanisms used	Phase compressed toward in-phase (0/45/90/135 degrees). Amplitude increased to maximum. Both-stroke AoA activated: recovery AoA raised to 15–18 degrees to generate lift on upstroke as well as downstroke. Stroke plane vertical.
Wave parameters	Phase: 0/45/90/135 degrees (more in-phase than hover). Amplitude: 65–70 degrees (maximum). Recovery AoA: 15–18 degrees (up from 5 degrees). All wings symmetric.
Expected result	Maximum instantaneous lift exceeds weight — aircraft climbs. Higher body oscillation than hover due to phase compression. Estimated climb rate: 1.4–2.8 m/s sustained, 3.5–5.5 m/s burst.

CONTROLLED DESCENT	
Mechanisms used	Counter-phase maintained but AoA on both strokes raised to similar values — both power and recovery strokes generate drag rather than efficient lift. Throttle reduced. The symmetric drag on both strokes slows descent rate.
Wave parameters	Phase: 0/180/0/180 degrees. AoA: 20 degrees both power and recovery strokes (versus 35/5 in hover). Amplitude: reduced to 45–50 degrees. Throttle: 40–55%.
Expected result	Controlled sink rate. Wings act as variable drag devices. Pitch attitude remains level. Estimated sink rate: 0.5–1.5 m/s.

CHAPTER 4 — TRANSLATIONAL MANEUVERS

LATERAL TRANSLATION — CRABBING	
Mechanisms used	Asymmetric stroke plane tilt: left-side wings stroke planes tilted rightward, right-side wings tilted leftward. Wave remains symmetric for stability. Sweep asymmetry adds lateral force component. No yaw input.
Wave parameters	Left wings: stroke plane tilted 10–15 degrees outboard (rightward). Right wings: stroke plane tilted 10–15 degrees inboard (leftward). Phase: symmetric 0/90/180/270 degrees both sides.
Expected result	Net force sideways with minimal yaw. Aircraft translates laterally while maintaining heading. Estimated crab speed: 0–4 km/h. Unique capability — no conventional fixed-wing aircraft can do this.

DIAGONAL TRANSLATION	
Mechanisms used	Combination of forward flight wave tilt PLUS lateral crab tilt. Both tilts applied simultaneously in software — no physical reconfiguration. The superposition of two tilt vectors produces a diagonal force vector.
Wave parameters	Forward component: 15 degree stroke plane forward tilt all wings. Lateral component: 8 degree asymmetric tilt left vs right. Phase: 0/90/180/270 degrees.
Expected result	Aircraft moves diagonally — forward and sideways simultaneously. Useful for precision positioning. Can reach any point in the horizontal plane by vector-summing forward and lateral components.

POSITION HOLD IN CROSSWIND	
Mechanisms used	IMU detects drift. Flight computer applies compensating lateral translation equal and opposite to measured drift velocity. Crab angle adjusts continuously to cancel wind drift. No pilot input required in GPS-hold mode.

Wave parameters	Dynamic crab angle computed from wind drift measurement. Lateral tilt magnitude proportional to drift velocity. Forward flight phase maintained for heading.
Expected result	Aircraft holds GPS position in crosswind up to 6 m/s. Heading maintained into wind to maximise gust rejection authority. Smooth automatic compensation with no attitude oscillation.

CHAPTER 5 — ROTATIONAL MANEUVERS AND TURNS

SPOT YAW — PIROUETTE	
Mechanisms used	Left side wave phase shifted forward by up to 180 degrees; right side phase shifted back by equal amount. This makes the left side generate predominantly forward thrust while the right side generates rearward thrust — a pure couple with zero net translational force. **This is the cleanest and most powerful yaw mechanism.**
Wave parameters	Left side phase: shifted +90 degrees (leads right side). Right side phase: shifted -90 degrees. Amplitude: symmetric. AoA: symmetric. Net translation: zero.
Expected result	Clean rotation about vertical axis. No translation. No altitude loss. Estimated maximum yaw rate: 45–80 degrees/second. Smoother than quadrotor yaw — no gyroscopic reaction torque.

BANKED TURN — COORDINATED	
Mechanisms used	Roll input (differential amplitude) produces bank angle. Simultaneously, yaw input (phase asymmetry in the direction of turn) keeps the turn coordinated by preventing sideslip. Altitude maintained by increasing overall amplitude to compensate for the reduced vertical lift component during the bank.
Wave parameters	Inside wing: amplitude reduced 20–30%. Outside wing: amplitude increased 20–30%. Yaw: phase shift toward inside of turn, 20–40 degrees. Throttle: increased 10–15% to maintain altitude in bank.
Expected result	Coordinated banked turn with no altitude loss and no sideslip. Similar feel to conventional fixed-wing coordinated turn. Bank angle proportional to roll input. Turn radius inversely proportional to bank.

FLAT TURN — SKID	
Mechanisms used	Yaw input only — no roll. Phase asymmetry left vs right generates yaw torque while amplitude remains symmetric on both sides. Aircraft yaws while maintaining level wings. Drag increase on the outside of the turn.
Wave parameters	Phase: left side leads, right side lags by 60–90 degrees. Amplitude: symmetric. Roll: zero. Altitude: maintained by throttle increase.
Expected result	Flat yaw with wings level — unusual in fixed-wing flight but natural to this architecture. Useful for heading changes in confined spaces where banked turns are not possible. Slower than banked turn for same heading change.

ROLL — AILERON EQUIVALENT	
Mechanisms used	Differential amplitude left vs right. Three-mechanism redundancy available: amplitude differential, AoA differential, and sweep differential all contribute. **Note: at 1 Hz, roll rate is modest compared to fixed-wing aircraft. Maximum approximately 35–65 degrees per second.**
Wave parameters	Roll left: left wing amplitude reduced 30%, right wing amplitude increased 30%. AoA augmentation: left wing AoA reduced 5 degrees, right raised 5 degrees. Sweep augmentation: right wings sweep slightly further forward.
Expected result	Roll moment generated. Aircraft banks in commanded direction. Rate proportional to input magnitude. Self-limiting — as bank increases, the effective moment arm reduces, giving natural rate limiting.

CHAPTER 6 — FIXED-WING AEROBATIC MANEUVERS

The following maneuvers are drawn from the fixed-wing aerobatic repertoire. Each is achievable with the metachronal cascade architecture, though the execution differs significantly from a conventional aircraft in several cases. The fundamental limitation is the 1 Hz cycle rate — attitude changes take 1–2 full cycles to establish, giving slower entry and exit than a fast fixed-wing. The fundamental advantage is that power and control authority are available at zero airspeed — maneuvers can be arrested and held at any point.

LOOP — INSIDE	
Mechanisms used	Entry in forward flight. Progressive nose-up pitch input (rear wing AoA increased, front wing AoA reduced, stroke plane tilted aft). As aircraft passes through inverted, AoA profile temporarily reversed for both-stroke lift. **Critical: symmetric airfoil (NACA 0015) on the symmetric+VG+covert variant supports inverted lift. S1223 does not.** Power maintained throughout.
Wave parameters	Entry: forward flight at cruise. Pitch input: rear wings +12 degrees AoA, front wings -8 degrees. Over the top: both-stroke AoA activated, amplitude increased. Exit: normal forward flight parameters restored.
Expected result	Circular loop in vertical plane. Diameter proportional to entry speed and pitch rate. Minimum loop diameter approximately 8–12 m at typical speeds. Altitude gain on pull-up, altitude loss on pull-out.

ROLL — AILERON ROLL	
Mechanisms used	Maximum roll input maintained for 360 degrees of bank. Simultaneously, pitch input applied to prevent nose drop during the inverted phase. Throttle increased to maintain altitude. The inverted phase requires both-stroke lift activation — recovery AoA raised to maintain positive lift when aircraft is upside down.

Wave parameters	Roll: maximum differential amplitude (30% left vs right). Pitch: nose-up compensation during inverted phase. Inverted: recovery AoA raised to 18–22 degrees both sides. Throttle: +20% throughout.
Expected result	Continuous roll. Rate limited by 1 Hz cycle — approximately 35–65 degrees/s. Full 360-degree roll takes approximately 6–10 seconds. Requires symmetric airfoil for efficient inverted phase.

SPLIT-S	
Mechanisms used	Half roll to inverted, then half loop pulling through to upright forward flight at lower altitude. Entry from forward flight. Roll to 180 degrees (inverted). Then nose-up pitch pull-through — in inverted flight this means pushing the flight path downward through a half circle back to upright.
Wave parameters	Phase 1 — half roll: differential amplitude to 180 degrees bank. Phase 2 — pull through: nose-up pitch input (inverted = pushing nose to sky reference, pulling through to upright). Both-stroke AoA during inverted. Phase 3 — exit: normal forward flight.
Expected result	Altitude loss of 15–30 m depending on entry speed. Heading reversal of 180 degrees. Useful for reversing direction while losing altitude. Entry speed should be low — this is a slow maneuver.

IMMELMANN TURN	
Mechanisms used	Half loop upward followed by half roll to upright — reverses direction while gaining altitude. Entry from forward flight at cruise. Full power nose-up pitch to climb through half loop. At apex (inverted), execute half roll to upright and exit in opposite direction to entry.
Wave parameters	Phase 1 — pull up: maximum nose-up pitch, full throttle. Phase 2 — over the top: both-stroke AoA, half roll input. Phase 3 — rollout: return to normal forward flight heading reversed.

Expected result	Altitude gain of 10–20 m. Heading reversal 180 degrees. Classic energy-gaining maneuver. Exit speed lower than entry — altitude traded for directional reversal.

HAMMERHEAD — STALL TURN	
Mechanisms used	Pull up to vertical, decelerate to near-zero airspeed, then yaw 180 degrees at the apex using differential wave phase, then dive away. Uniquely suited to this aircraft because hover capability means the apex can be held indefinitely at zero airspeed — no minimum speed constraint.
Wave parameters	Phase 1 — pull up: maximum climb, full throttle. Phase 2 — apex: transition to hover mode. Aircraft stationary, vertical. Phase 3 — yaw: full yaw input (differential wave phase), 180 degrees. Phase 4 — dive: nose-down pitch, transition to forward flight.
Expected result	Classic hammerhead with the unique advantage that the hover phase can be held at the apex. Dramatic maneuver. Altitude at apex equals pull-up height. Yaw rate at apex: 45–80 degrees/second.

KNIFE-EDGE FLIGHT	
Mechanisms used	Aircraft rolled to 90 degrees — wings vertical. Lift vector now horizontal. Altitude maintained by generating lateral thrust — left-side wings generate upward thrust (now pointing up), right-side wings generate downward thrust. This requires differential stroke plane tilt — left side tilted to push up, right side to push down.
Wave parameters	Bank: 90 degrees (wings vertical). Left wings: stroke plane tilted upward — generates lift in vertical direction. Right wings: stroke plane tilted downward — counter-torques the roll. Forward speed: maintained by normal forward wave tilt.
Expected result	Sustained knife-edge flight. Altitude maintained by stroke plane asymmetry. Limited duration — control authority is reduced in this attitude. Not natural to this aircraft but achievable briefly.

SNAP ROLL — FLICK ROLL	
Mechanisms used	Rapid autorotation around the roll axis at high AoA — one wing stalls while the other continues generating lift. Applied by sudden full roll input combined with high pitch input (both wings at high AoA simultaneously). One side stalls first due to roll-induced AoA increase on the retreating wing.
Wave parameters	Simultaneous inputs: maximum roll (differential amplitude) PLUS maximum pitch up (high AoA all wings). The advancing wing stalls, the retreating wing continues flying — autorotation begins.
Expected result	Rapid snap roll — faster than aileron roll due to autorotative augmentation. Self-limiting: aircraft exits the snap naturally as AoA reduces. Requires practice to stop at desired bank angle.

CHAPTER 7 — ROTARY-WING MANEUVERS

The metachronal cascade architecture bridges fixed-wing and rotary-wing capabilities. The following maneuvers are drawn from the helicopter and multirotor repertoire and are generally more natural to this aircraft than the fixed-wing aerobatics above, owing to the VTOL and hover capability.

PEDAL TURN — IN-PLACE PIROUETTE	
Mechanisms used	Pure yaw with zero translation. The most natural and effective maneuver for this architecture. Differential wave phase creates opposing thrust vectors left and right — a pure couple. The aircraft spins about its vertical axis with no tendency to translate. Equivalent to helicopter pedal input.
Wave parameters	Left side: wave phase advanced by 90 degrees. Right side: wave phase retarded by 90 degrees. Amplitude: symmetric. Stroke plane: horizontal, no tilt. Throttle: maintained at hover setting.
Expected result	Clean rotation. Rate: 45–80 degrees/second maximum. 360-degree turn in approximately 5–8 seconds. Zero translation. Zero altitude change.

QUICK STOP — RAPID DECELERATION	
Mechanisms used	Rapid arrest from forward flight. Pitch nose-up sharply — all wings tilted to generate rearward force vector. Simultaneously, all wings switched to maximum drag AoA on both strokes — both power and recovery stroke generate braking drag. Dynamic stall used as airbrake at higher entry speeds.
Wave parameters	All wings: both-stroke AoA raised to 38–42 degrees (drag mode). Stroke plane: tilted aft 30–40 degrees (thrust vector reversed). Amplitude: maximum. Phase: counter-phase for maximum cycle frequency.

Expected result	Maximum deceleration up to approximately 6–8 m/s squared. Forward speed arrested within 2–4 m from cruise speed. Aircraft pitches nose-up then settles to hover. Similar to swift post-stall arrest maneuver.

HOVER TAXI	
Mechanisms used	Very slow translational flight at constant altitude — 0.5 to 3 m/s. Small forward tilt of stroke plane — just enough to produce gradual translation. IMU closed loop maintains altitude and heading. GPS hold maintains track if available.
Wave parameters	Stroke plane: 3–8 degree forward tilt. Altitude: IMU feedback loop. Heading: yaw hold. Speed: proportional to stroke plane tilt angle.
Expected result	Precise hover taxi. Predictable speed control. Highly effective for inspection, perching approach, or navigation in confined spaces.

NOSE-IN HOVER	
Mechanisms used	Hover with aircraft pointing toward the pilot — most disorienting orientation for manual control. All control inputs are reversed from pilot perspective. Flight controller handles this transparently — pilot inputs are in world frame, firmware translates to aircraft frame regardless of heading.
Wave parameters	Identical to normal hover — wave parameters unchanged. Flight controller uses IMU heading to rotate pilot input vector to aircraft body frame. No wave modification required.
Expected result	Stable hover in any heading orientation. Pilot controls in world frame throughout. Aircraft heading is independent of control authority.

FUNNEL — ORBIT

Mechanisms used	Aircraft orbits a fixed point in space while maintaining a fixed attitude (nose always pointing outward, or inward, or at fixed heading). Achieved by continuously varying the stroke plane tilt direction to point toward the orbit centre while maintaining altitude.
Wave parameters	Stroke plane tilt: continuously rotated to point at orbit centre. Magnitude: proportional to desired orbit speed. Yaw: varies continuously to maintain desired heading reference. Altitude: maintained by throttle feedback.
Expected result	Circular orbit around a point. Radius set by stroke plane tilt magnitude. Can maintain nose-in, nose-out, or fixed heading during orbit. Useful for FPV inspection of a central subject.

TOILET BOWL — CORRECTION DEMONSTRATION	
Mechanisms used	Uncommanded slow orbit caused by slight yaw trim error combined with forward drift. The aircraft gradually circles rather than hovering in place. Corrected by trimming yaw to zero and engaging GPS position hold. Understanding this failure mode is important for initial flight testing.
Wave parameters	Cause: yaw trim error combined with forward drift. Correction: yaw trim to zero; reduce forward tilt trim to zero; engage position hold.
Expected result	This is a diagnostic mode, not a commanded maneuver. Inclusion here is as a practical warning for initial flight testing. Correcting this tendency is an early calibration task.

CHAPTER 8 — EMERGENCY AND RECOVERY MANEUVERS

EMERGENCY GLIDE — POWER FAILURE	
Mechanisms used	On power loss, flight computer immediately commands all wings to sweep to fixed position at optimal glide AoA. Wings become fixed lifting surfaces. Battery backup powers servos for minimum 5 minutes in this mode. Metachronal wave stops — all wings locked at 12 degrees AoA, forward sweep position.
Wave parameters	All wing stroke servos: commanded to neutral (0 degrees stroke). All wing pitch servos: held at 12 degrees AoA. All wing sweep servos: held at 10 degrees forward sweep. Battery: backup powers servos only — approximately 8W.
Expected result	Glide ratio 8:1 to 12:1. Aircraft descends on gentle glide path. No control of heading or bank — emergency landing wherever glide path leads. Better outcome than uncontrolled descent.

INVERTED RECOVERY	
Mechanisms used	Aircraft has been inverted by gust or upset. IMU detects bank exceeding 90 degrees. Automatic recovery sequence: maximum roll input toward upright, both-stroke AoA to prevent altitude loss during recovery, throttle increase to maintain altitude.
Wave parameters	Roll: maximum roll input toward nearest upright path. AoA: both-stroke mode activated (recovery AoA raised to 18 degrees). Throttle: +25% during recovery. Pitch: neutral during roll, then nose-up correction as upright approached.
Expected result	Recovery from inverted in approximately 3–5 seconds depending on bank angle. **Note: recovery from fully inverted requires symmetric airfoil variant. S1223 has limited inverted lift. The symmetric+VG+covert configuration is recommended for aerobatic operations.**

WING FAULT RECOVERY	
Mechanisms used	One wing fails (servo fault, structural failure). IMU detects roll/yaw divergence. Firmware identifies failed wing from servo current monitoring. Automatically stows failed wing, recalculates phase spacing for remaining seven wings, applies compensating amplitude increase on mirror wing.
Wave parameters	Failed wing: stroke and pitch servos commanded to neutral, sweep to full stow. Mirror wing: amplitude increased by 15–20% to compensate. Phase: redistributed evenly across remaining 7 active wings. Roll/yaw correction: applied via amplitude and AoA asymmetry.
Expected result	Continued controlled flight with 7 of 8 wings. Some degradation in efficiency and maneuver authority. Land as soon as practical. Minimum operable configuration: 6 of 8 wings.

SPIN RECOVERY	
Mechanisms used	Uncommanded autorotation — aircraft spinning uncontrollably. Counter-yaw at maximum authority (full differential phase). Simultaneously, all wings to minimum AoA to reduce any stalled surface, then throttle reduced slightly to reduce gyroscopic effects.
Wave parameters	Yaw: maximum counter-spin differential phase input. AoA: all wings reduced to 15 degrees (reduce stall tendency). Throttle: reduce to 60% during recovery. Roll: neutral — let yaw recovery dominate.
Expected result	Spin arrested in 1–3 revolutions depending on entry rate. Similar to spin recovery in conventional aircraft — counter-rudder is the primary input, all others secondary.

CHAPTER 9 — WAVE PARAMETER QUICK REFERENCE

The following table summarises the primary wave parameters for every flight mode covered in this document. Parameters marked with an asterisk (*) are managed automatically by the flight control firmware in response to RC stick inputs — the pilot does not need to set these manually.

Maneuver	Phase (W1-W4)	Amplitude	AoA	Sweep	Power
Stationary hover	0/180/0/180	Symmetric	35/5 both	Mod fwd/aft	Low
Forward cruise	0/90/180/270	Symmetric	28/3 both	30 fwd stroke	Low
Reverse flight	270/180/90/0	Symmetric	28/3 both	Reversed	Low
Vertical climb	0/45/90/135	Symmetric	40/15 both	Mod fwd	High
Descent	0/180/0/180	Symmetric	20/20 both	Moderate	Low
Lateral crab	0/90/180/270	Side tilt	28/5 both	Standard	Low
Roll left	0/90/180/270	L low R hi	Asymm L low	L less fwd	Low
Pitch up	0/90/180/270	Symmetric	Rear hi/Fnt lo	Standard	Low
Yaw left	L leads R	Symmetric	Symmetric	Standard	Low
Banked turn L	0/90/180/270	L low R hi	Symmetric	Standard	Med
Flat turn L	L leads R	Symmetric	Symmetric	Standard	Low

Maneuver	Phase (W1-W4)	Amplitude	AoA	Sweep	Power
Loop (inside)	0/90/180/270	Symmetric	Rear increase	Standard	High
Aileron roll	0/90/180/270	Max diff	Inverted comp	Standard	High
Split-S	0/90/180/270	Half roll	Both-stroke	Varies	High
Immelmann	0/90/180/270	Symmetric	Nose up max	Standard	High
Hammerhead	Hover at top	Symmetric	Max climb	Hover at top	High
Snap roll	0/90/180/270	Max diff	Max ALL wings	Standard	High
Quick stop	0/180/0/180	Symmetric	38/38 drag	Aft tilt	High
Emergency glide	ALL STOPPED	Wings locked	12 deg fixed	Fwd fixed	None
Wing fault recovery	7-wing redist	Compensated	Compensated	Standard	Med

IMPORTANT: All maneuver entries above describe the primary wave modification. In practice, the flight computer blends between modes smoothly over 2–3 wing cycles to avoid abrupt transitions. The pilot inputs on the RC transmitter map directly to conventional aircraft controls (stick, throttle, rudder). The firmware performs all wave parameter translations automatically. The pilot does not interact with wave parameters directly during normal flight.

Understanding the Limits

The fundamental limit of this aircraft for aerobatic performance is the 1 Hz flap cycle. At one cycle per second, attitude changes take 1–2 seconds to establish — comparable to a very large slow aircraft. This means entries and exits to maneuvers

are deliberate and gradual. What the architecture gains in exchange is unique: the ability to arrest any maneuver at any point and transition to a stable hover. A loop can be stopped halfway. A roll can be frozen at 90 degrees. A hammerhead can be held at the apex indefinitely. No conventional aircraft — fixed-wing or rotary — can do this. The combination of slow graceful aerobatics with instant hover recovery at any point defines a genuinely new flight character.

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